

MIRAGE-X

Integrated Adaptive Cyber Defense & Attacker Intelligence Platform

Ignition × United Hackathon 2026

Cybersecurity Concept • Hackathon MVP → Final-Year Project

Core vision: MIRAGE-X combines behavioral attack profiling, attack-path intelligence, adaptive cyber deception, incident memory, and explainable AI into one defensive platform. The goal is not simply to detect an attacker, but to understand the attack, predict where it may go, dynamically change the defensive environment, and turn attacker activity into useful intelligence.

1. Why Combine Multiple Projects?

The strongest version of this project is not a collection of unrelated cybersecurity tools. Each component solves one part of the same problem: **how to understand and contain an evolving attacker**. The components form a single closed-loop defense system.

Attack Sensor → Attack DNA → Attack Path → Adaptive Deception → Cyber Memory → AI Defender

2. Unified Architecture

```
ATTACKER / CONTROLLED SIMULATOR
↓
1. Attack Sensor – logs, network events, authentication events
↓
2. Attack DNA – behavioral sequence and profile
↓
3. Attack-Path & Intent Engine – likely objective and next target
↓
4. Adaptive Deception Engine – selects controlled decoy
↓
Fake SSH • Fake Database • Fake Admin Panel
↓
5. Cyber Incident Memory – stores attack patterns and outcomes
↓
6. Explainable AI Defender – summarizes, correlates and recommends defensive actions
↓
SECURITY DASHBOARD
```

3. Component Breakdown

Component	Role	Hackathon MVP
Attack Sensor	Collect and normalize controlled security events.	SSH/auth/request logs
Attack DNA	Represent behavior rather than relying only on IP/hash indicators.	Sequence + risk features
Attack-Path Intelligence	Estimate where the attacker may move next.	Small simulated network graph
Adaptive Deception	Choose a decoy according to observed behavior.	2–3 decoy services
Cyber Memory	Store previous incidents and compare new behavior.	Similarity + incident history
AI Defender	Explain what happened and recommend defensive action.	Human-readable incident summary
Dashboard	Present the whole attack story visually.	Timeline + graph + alerts

4. Example Attack Scenario

Use a controlled attacker simulator so the entire demo is repeatable and safe. The attacker starts with a reconnaissance/authentication phase and progressively interacts with the simulated environment.

```
Reconnaissance
↓
SSH scanning
↓
Credential attempts
↓
Successful test login
↓
Privilege-related command
↓
Suspicious download
↓
Search for database/admin resources
```

MIRAGE-X response: the system builds Attack DNA, estimates the likely attack path, increases the risk score, and changes the available deception. If the attacker starts searching for database assets, the controller can expose a controlled fake database instead of exposing any real resource.

Result: the attacker remains inside an isolated deception environment while the system collects telemetry about commands, requests, timing, movement, and interaction choices.

5. The Unique System-Level Idea

The project should not be marketed as 'five projects combined'. The central idea is an **adaptive defensive loop**: the system uses attacker behavior to decide what defensive environment the attacker should see next.

Instead of only asking:

“Is this attack malicious?”

MIRAGE-X asks:

“What is this attacker trying to accomplish, what will they likely do next, and which controlled deception should we deploy now?”

6. Hackathon MVP — Keep It Small

Priority	Feature	Goal
P0	Controlled attack simulator	Create repeatable attack behavior.
P0	Attack DNA	Convert events into behavioral profile.
P0	Adaptive deception	Switch between 2–3 decoys.
P0	Dashboard	Show live attack story.
P1	Incident memory	Compare with previous simulated attacks.
P1	AI explanation	Explain attack and recommendation.

7. Final-Year Project Expansion

Phase	Expansion
MIRAGE-X v1	Hackathon MVP with simulator, Attack DNA, adaptive decoys and dashboard.
v2	Real lab telemetry, richer attack graphs, incident correlation and threat-intelligence enrichment.
v3	Machine-learning behavior models, campaign clustering, deception strategy optimization and explainability.
Research	Measure detection latency, false positives, attacker engagement, decoy utility and intelligence quality.
Advanced	Optional AI-agent security and digital-twin simulation as separate modules.

8. Recommended Technology Stack

Backend: Python + FastAPI
Frontend: React + Vite
Data: SQLite for MVP → PostgreSQL later
Isolation: Docker containers
Analysis: Python rules + optional ML models
Visualization: attack graph, event timeline, risk indicators
Security mapping: MITRE ATT&CK; references where appropriate

9. 13-Day Build Strategy

Days	Deliverable
1–2	Architecture, threat model, UI wireframe, attack scenario.
3–4	Event collector + attacker simulator.
5–6	Attack DNA + risk engine.
7–8	Fake SSH/database/admin decoys + interaction logging.
9–10	Adaptive deception controller + attack-path visualization.
11	Incident memory + AI explanation.
12	Dashboard integration, testing and demo reset.
13	Pitch, documentation, screenshots, rehearsal and final fixes.

10. Final-Year Research Question

Can an adaptive cyber-defense platform use attacker behavioral profiles and predicted attack paths to dynamically select deception strategies, increase useful attacker engagement, and generate actionable defensive intelligence?

11. Evaluation Metrics

- Detection latency
- Attack classification accuracy
- False-positive rate
- Attack-DNA similarity quality
- Attack-path prediction quality
- Decoy-selection utility
- Attacker engagement time in the isolated environment

- Quantity and quality of collected intelligence
- Explainability and analyst usefulness

12. Safety Boundary

All demonstrations should run in an isolated, authorized laboratory environment. Use a controlled attacker simulator or deliberately generated test traffic. Decoys must not contain real credentials, sensitive data, or uncontrolled access to external systems.

13. One-Line Hackathon Pitch

“MIRAGE-X doesn't just detect attackers — it understands their behavior, predicts their path, dynamically changes the environment they see, and turns their actions into defensive intelligence.”

14. Important Novelty Note

The project should not claim that every individual technique is new. Honeypots, attack graphs, behavioral analysis, incident correlation and AI-assisted security are established research areas. The defensible contribution is the specific integrated architecture, adaptive decision mechanism, implementation, and measured results produced by the project.